

Computer Network

Cryptography In class Assignment

Only written answers are accepted.

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Marks allocated: 20%

1. Define Cryptography and explain why it is necessary in network communication.
2. List and briefly explain any three security services provided by cryptography.
3. Differentiate between Plaintext and Ciphertext with an example.
4. What is a cryptographic key? Why is key secrecy more important than algorithm secrecy?
5. Define Encryption and Decryption.
6. Differentiate between Substitution Cipher and Transposition Cipher (minimum 3 points).
7. Compare Symmetric Key Cryptography and Asymmetric Key Cryptography using the following aspects:
 - Number of keys
 - Speed
 - Key distribution
8. Explain the difference between Block Ciphers and Stream Ciphers.
9. Why are symmetric encryption algorithms called bulk ciphers?
10. Draw and explain the process of Symmetric Encryption, clearly showing:
 - Plaintext
 - Encryption Algorithm
 - Secret Key
 - Ciphertext
11. Draw and explain the Public Key Encryption process for communication between Alice and Bob.
12. Briefly explain why asymmetric encryption is important for secure key distribution.
13. Differentiate between DES, 3DES, and AES based on:
 - Key size
 - Security level
 - Current usage
14. Why was AES introduced to replace DES and 3DES? Mention at least two reasons.

15. Explain the working principle of Digital Signatures, highlighting:

- Role of hashing
- Authentication
- Integrity
- Non-repudiation

16. What is a Digital Certificate? List any four pieces of information contained in a digital certificate.

17. Explain how Public Key Infrastructure (PKI) helps in:

- a. Certificate validation
- b. Trust establishment